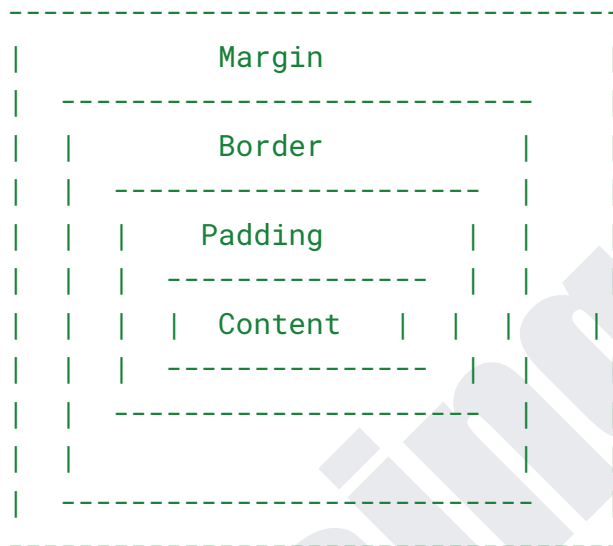


📌 1. What is the CSS Box Model?

The **CSS Box Model** describes how elements are displayed as rectangular boxes on a webpage. Every HTML element is treated as a **box**, and the Box Model consists of four key parts:

- ❶ **Content** – The actual text, image, or element inside the box.
- ❷ **Padding** – The space between the content and the border.
- ❸ **Border** – A visible boundary around the padding and content.
- ❹ **Margin** – The space outside the border, separating elements from each other.

📌 **Diagram of Box Model:**



♦ Live Demo (Example HTML + CSS Box Model)

html

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```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>CSS Box Model</title>
  <style>
    .box {
      width: 300px;
```

```
        height: 150px;
        background-color: lightblue;
        padding: 20px;
        border: 5px solid blue;
        margin: 30px;
    }
</style>
</head>
<body>
    <div class="box">This is a box</div>
</body>
</html>
```

Breakdown of Code:

- ✓ width: 300px; height: 150px; → The content area size.
 - ✓ padding: 20px; → Space inside the box, between content and border.
 - ✓ border: 5px solid blue; → The boundary of the box.
 - ✓ margin: 30px; → Space between this box and other elements.
-

2. Margin – Space Outside the Border

Margin creates space outside an element's border, pushing it away from surrounding elements.

◆ Margin Syntax:

css

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```
margin: 20px; /* Applies 20px space on all sides */
margin-top: 10px;
margin-right: 15px;
margin-bottom: 20px;
margin-left: 25px;
```

◆ Live Demo (Margin Example)

css

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```
.box {
    margin: 50px; /* Adds 50px space outside the border */
}
```

Margin Collapse

Sometimes, the margins of two elements collapse into one instead of adding up.

Example:

- If an `<h1>` has `margin-bottom: 20px;` and a `<p>` below it has `margin-top: 30px;`, instead of **50px total**, it will only take the **larger margin (30px)**.
-

3. Padding – Space Inside the Border

Padding adds space inside the border but outside the content.

♦ Padding Syntax:

css

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```
padding: 20px; /* Applies 20px padding on all sides */
padding-top: 10px;
padding-right: 15px;
padding-bottom: 20px;
padding-left: 25px;
```

♦ Live Demo (Padding Example)

css

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```
.box {
    padding: 20px; /* Adds 20px space inside the box */
}
```

♦ Difference Between Margin & Padding:

Property	Effect
Margin	Adds space outside the border
Padding	Adds space inside the border

4. Border – The Edge Around an Element

A **border** wraps around the padding and content, creating a visible boundary.

◆ Border Syntax:

```
css
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border: 3px solid red;
border-width: 5px;
border-style:    ;
border-color: green;
```

◆ Border Styles:

Style	Example
-------	---------

solid	
-------	---

dashed	
--------	---

dotted	
--------	---

double	
--------	---

◆ Live Demo (Border Example)

```
css
CopyEdit
.box {
  border: 5px dashed green;
}
```

5. Width & Height – Controlling Box Size

The **width** and **height** properties define the size of the content area (**excluding padding, border, margin**).

◆ Width & Height Syntax:

```
css
```

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width: 400px;
height: 200px;
max-width: 100%; /* Ensures element never exceeds screen width */
min-height: 100px; /* Sets minimum height */

♦ **Live Demo (Width & Height Example)**

CSS
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.box {
 width: 250px;
 height: 150px;
}

6. Overflow – Handling Extra Content

Sometimes, content overflows the box, and you need to control how it's displayed.

♦ **Overflow Values:**

Value	Effect
visible	Default, content overflows
hidden	Cuts off content
scroll	Adds scrollbars
auto	Adds scrollbars only when needed

♦ **Live Demo (Overflow Example)**

CSS
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.box {
 width: 200px;
 height: 100px;
 overflow: scroll;
}

}

Presentation Tips

- ✓ Use **visuals and diagrams** to explain the Box Model.
 - ✓ Show **before/after changes** live using CSS code.
 - ✓ Engage students by making them try out **margin**, **padding**, and **border**.
 - ✓ Ask **questions**, e.g., "What happens when we increase padding? What if margin is negative?"
-

Summary

- ✓ **Margin** = Space outside the border.
- ✓ **Padding** = Space inside the border, around content.
- ✓ **Border** = The boundary around the element.
- ✓ **Width & Height** = Controls the size of the box.
- ✓ **Overflow** = Manages extra content.